Water consumption and sustainability of avocado cultivation in the municipality of Patzcuaro, Michoacan, Mexico

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1. Introduction

The avocado is a tropical fruit native to Central America. Mexico is the world's largest producer, accounting for 28.4% of global production [1]. **Michoacan is Mexico's largest avocado-producing state, accounting for 76% of national productio**n [2]. The expansion of avocado cultivation has contributed significantly to Michoacán's economy [3,4]. However, this growth has also created negative social and environmental impacts, including land-use changes, the appropriation of rural lands, biodiversity loss, and increased water retention for irrigation [5, 6, 7, 8, 9, 10].

In Michoacan, avocado cultivation has expanded rapidly in the municipality of **Patzcuaro, increasing from 8 hectares in 2003 to 1,260 hectares in 2024** [2]. This expansion has reshaped local agriculture, shifting from traditional farming towards avocado monoculture (Figure 1) [11]. **It has also contributed to environmental degradation**, particularly affecting Lake Patzcuaro, which declined to less than 47% of its capacity between 2023 and 2024. Moreover, increasing use of water reservoirs for avocado irrigation has placed additional pressure on local water resources, limiting water availability for some communities [12,13]. **This study examines how water consumption for avocado cultivation changed between 2003 and 2024**, and its implications for water sustainability in Patzcuaro.

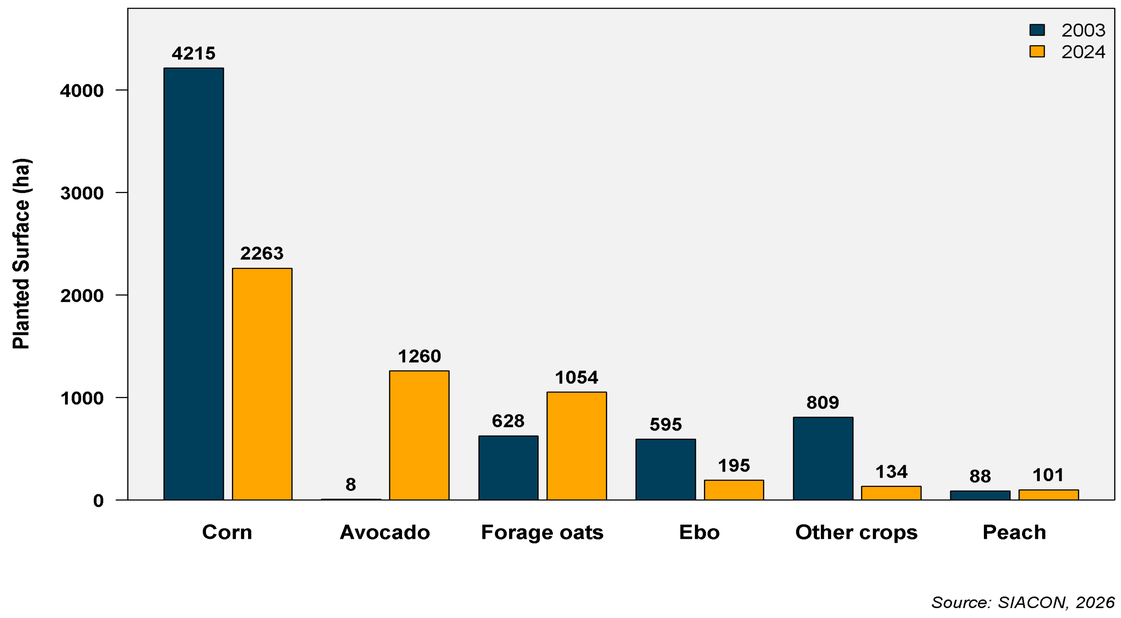


Figure 1. Planted Surface by Crop (2003 vs. 2024)

2. Study area

The municipality of **Patzcuaro is located in central Michoacán, within the Purépecha region,** at an altitude ranging from 2,100 to 3,300 meters above sea level (Figure 2). Patzcuaro has a temperate, sub-humid climate characterized by summer rainfall. Avocado plantations cover approximately 4,294 hectares, accounting for 10% of the municipality's total area [14].

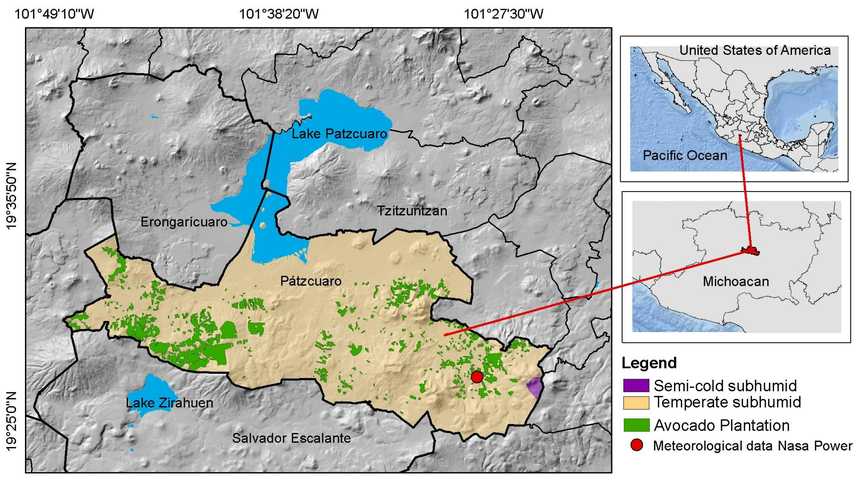


Figure 2. Geographic location of Patzcuaro and avocado plantations

3. Method

To estimate changes in water consumption between 2003 and 2024, I calculated crop **water requirements (CWR) using CROPWAT 8.0 software** [15]. CWR represents the amount of water required for optimal crop growth and is estimated based on crop evapotranspiration (ETc), calculated as CWR = ETo × Kc, where ETo is the reference evapotranspiration, and Kc is the crop coefficient [15].

To assess water sustainability, I calculated a **sustainability indicator (SI)** as the ratio of annual blue water consumption (irrigation) to the volume of water allocated for agricultural use in the municipality, expressed as a percentage. This indicator measures whether estimated irrigation water consumption exceeds the allocated volume.

The **SI provides a conservative measure of pressure on allocated water resources**, as it assumes that the entire agricultural water allocation is available for avocado cultivation, without accounting for water demand from other crops. An SI above 100% means that estimated irrigation water consumption exceeds the total agricultural water allocation, even under this assumption. However, values of 100% or below do not necessarily imply sustainable water use, as the indicator does not account for competing water demands, ecological water requirements, or the long-term availability of water resources.

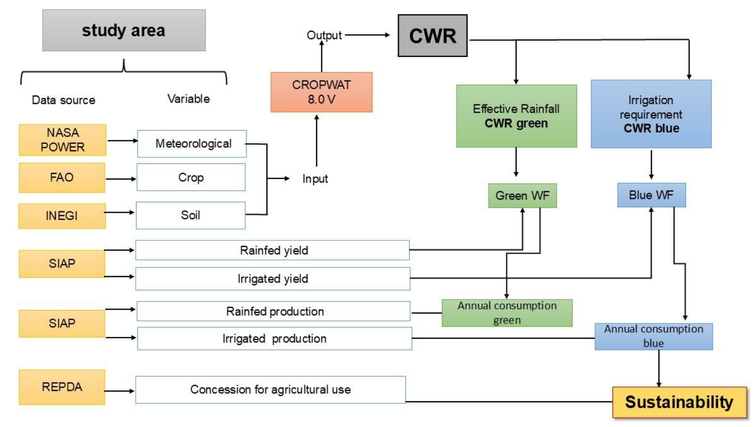


Figure 3. Diagram of the procedure used to analyze the green and blue water footprints

4. Results and discussion

Between 2003 and 2024, **avocado production increased substantially, from 50 to 12,807.5 tons, a more than 255-fold increase.** Over the same period, estimated **blue water consumption (irrigation) increased from 0.06 hm³ to 8.58 hm³**, approximately 143 times its initial level. In contrast, the volume of water allocated for agricultural use increased from 0.17 to 0.86 hm³, **approximately five times its initial level. As a result, the sustainability indicator (SI) increased from 35% in 2003 to 994% in 2024**. This indicates that estimated irrigation water consumption for avocado cultivation in 2024 was almost ten times the total agricultural water allocation, even under the conservative assumption that the entire allocated volume was available for avocado cultivation (Figure 4).

Similar pressures on water resources have been documented in other avocado-producing regions. In Petorca and La Ligua, central Chile, avocado cultivation has been associated with the depletion of water resources available to rural communities, leading to social protests and boycotts of Chilean avocados in some European countries [16]. In Mexico, Gómez-Tagle et al. [10] found that water demand from avocado cultivation in Uruapan reached up to 120% of the allocated volume during dry years. Similarly, Fuerte-Velázquez and Gómez-Tagle [9] found that water demand in Ziracuaretiro was 27.7% higher than in neighboring Uruapan.

These findings highlight the importance of improving water allocation and **monitoring mechanisms, including water footprint assessments, to manage increasing pressure on local water resources.**

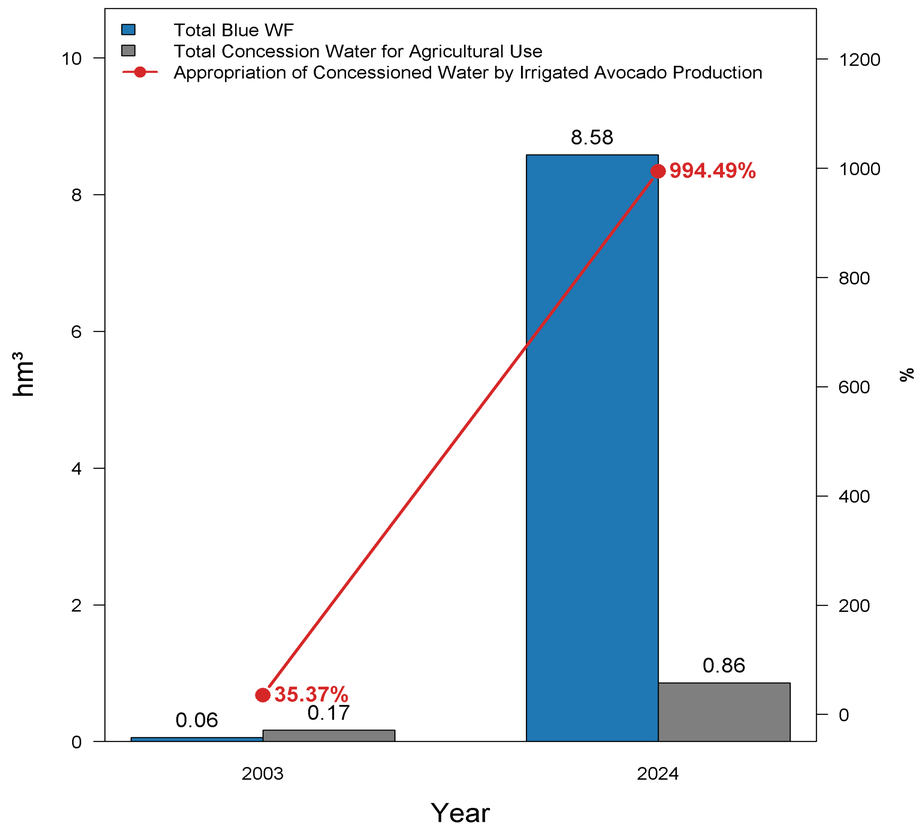


Figure 4. Sustainability indicator (SI)

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